

NYC Airbnb Compliance Analysis

Data Science and AI for Business TECH-UB 57 Spring 2026

Group A4

Jiaqi Zhu (jz6553)

Tanzim Ali (ta2812)

Josh Kwon (bk2351)

Ziyu Tian (zt2382)

Link to Raw Data: [Inside Airbnb](#)

Link to Python Notebook: [Python Notebook](#)

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1. Business Understanding

1.1 Problem Context and Motivation

New York City has long struggled with housing affordability, and conditions have worsened in recent years. The city's rental vacancy rate fell to just 1.4% in 2023, which is a multi-decade low, reflecting a market under severe strain (NYC Comptroller's Office). A contributing factor is the conversion of residential apartments into short-term rentals on platforms like Airbnb, which research has linked to reduced long-term housing supply and upward rent pressure in affected neighborhoods (Soltas). In response, New York City enacted Local Law 18 (LL18) in September 2023, requiring hosts to register with the Mayor's Office of Special Enforcement (OSE) and prohibiting entire-apartment rentals of fewer than 30 nights without the host present.

Yet enforcement has remained largely reactive, and a deeper problem persists: sophisticated hosts may manipulate key listing fields, such as minimum night settings or licence disclosures, to appear compliant while continuing to operate illegally. For example in May 2025, NYC filed its first LL18 lawsuit against an operator caught “misrepresenting their lawful status to online booking websites to evade the registration law's prohibitions” (NYC Mayor's Office of Criminal Justice). These surface-compliant but genuinely illegal listings, which we term “Shadow Listings”, cannot be identified by rule-based checks alone. This project addresses that gap by training a machine learning model on several features to surface them and produce a risk-ranked audit list for OSE field inspectors.

1.2 Business Problem Definition

Big Problem: How can we efficiently identify potential illegal short-term rental operations among tens of thousands of active listings to protect the housing market?

Small Problem: What is the probability that a specific listing is operating illegally under LL18, and which listing features are the strongest predictors of non-compliance?

1.3 Role and Target Audience

Our team operates as Data Analysts within the NYC OSE. This report is addressed to the Executive Director of the OSE and the NYC Housing Commissioner. The intended outcome is a ranked list of high-risk listings that field inspectors can use to prioritize site visits, increasing the department's hit rate per inspection and delivering measurable public value through fine revenue

and housing market stabilization. Shifting from reactive to proactive enforcement represents not only a financial improvement but a structural change in the department's relationship with LL18 compliance.

2. Data Understanding

2.1 Data Source

The dataset was sourced from Inside Airbnb, a non-commercial project that publishes scraped Airbnb listing data for major cities worldwide. The specific data used was collected on October 1, 2025 and covers all active Airbnb listings within New York City. Inside Airbnb collects data via automated scraping of the public Airbnb website, meaning the dataset reflects information that hosts choose to display publicly and is not independently verified by any regulatory authority. This characteristic of the data is both a limitation and a methodological motivation.

2.2 Dataset Overview

The raw dataset comprises 36,111 rows and 79 columns. Each row represents a unique active Airbnb listing. Columns span a broad range of attributes including host-level characteristics, property attributes, pricing and availability metrics, review scores, license registration information, and geographic coordinates. Data quality varies considerably across columns.

2.3 Potential Bias and Limitations

Several sources of bias need to be acknowledged before proceeding to analysis. First, the Inside Airbnb data reflects only listings active at the time of scraping, excluding previously suspended or de-listed properties. Second, hosts aware of LL18 may strategically manipulate key listing features to appear compliant while operating differently in practice -- a form of fake compliance that our model is specifically designed to detect. Third, the license field is sparsely populated: only 5,376 of 36,111 listings (14.9%) carry any license entry, making it difficult to distinguish between genuinely unlicensed hosts and exempt hosts who have not completed the field.

3. Data Preparation

3.1 Feature Selection

From the original 79 columns, 20 features were selected based on their relevance to LL18's legal requirements, host commercial behavior, and listing activity. Identifier fields, text-heavy

columns (e.g., listing descriptions), and features with unimportant information were excluded. The selected features are summarized in the Table below:

Category	Feature(s)	Rationale
Identification & Metadata	id — Unique identifier for the listing host_id — Unique identifier for the host	Used for record linkage and deployment output; not used as model features.
*Legal Logic (used to construct is_illegal, then dropped)	room_type — To identify “Entire home/apt” listings minimum_nights — To identify stays < 30 days license — OSE registration status	Directly encode LL18's two compliance criteria. Dropped before modeling to prevent data leakage.
Host Professionalism & Commerciality	host_acceptance_rate — High rates suggest revenue maximization host_is_superhost — Indicates high service standards for commercial gain host_neighbourhood — Host's self-reported location calculated_host_listings_count_entire_homes — Specific count of high-risk “entire home” units calculated_host_listings_count_private_rooms — Count of private room inventory	High values indicate professional, multi-unit operations rather than casual home-sharing.
Pricing & Economic Motivation	price — Signals illegal short-term premiums estimated_revenue_1365d — Potential ROI driving non-compliance	Elevated pricing and annual revenue signal commercial intent and financial motivation for non-compliance.
Activity & Turnover	availability_365 — Measures if the unit is a dedicated business number_of_reviews_ltm — Review frequency indicates guest turnover number_of_reviews_130d — Very recent activity	High year-round availability and review frequency are hallmarks of dedicated short-term rental operations.
Location & Market Hotspots	neighbourhood_cleansed — Specific neighborhood for community impact neighbourhood_group_cleansed — The Borough (5 in total; one-hot encoded)	Enable borough-level geographic segmentation and spatial mapping for OSE deployment.
Operational Features	instant_bookable — Used by operators to automate high volume	Automated booking is common among high-volume commercial operators.
Location on Map	latitude — For mapping and spatial analysis longitude — For mapping and spatial analysis	Geographic coordinates used in deployment output for inspector dispatch.

Table 1: Selected Features by Category

3.2 Handling Missing Values (With Some Transformations)

Missing values were addressed through context-aware strategies rather than uniform imputation:

- **price:** Converted from currency strings to numeric. Missing values are imputed using the median price within each listing's specific neighbourhood, respecting the significant geographical price variations across NYC, with a global median fallback for neighbourhoods with no price data.
- **host_acceptance_rate:** Converted from percentage strings to floats. Rather than dropping rows, a binary indicator feature `has_acceptance_rate` was engineered (1 if present, 0 if missing), preserving the informational value of non-disclosure. The rate itself was then imputed using the within-group median stratified by `host_is_superhost`, since acceptance behavior differs systematically between superhosts and regular hosts. Evidence will be shown in the EDA part.
- **estimated_revenue_1365d:** Missing values filled with 0, reflecting zero observed booking revenue.
- **host_is_superhost:** Rare missing values are imputed with the mode ('f'), then mapped to binary integers (1 = superhost, 0 = otherwise).

3.3 Encoding and Transformations

Boolean strings and categorical fields were encoded into numeric representations to prepare the dataset for modeling:

- **instant_bookable:** Mapped from boolean strings to binary integers.
- **neighbourhood_group_cleansed:** Encoded into five borough dummy variables, enabling borough-level signal capture in following models.
- **license:** All entries containing “ose” (case-insensitive) were standardized to “OSE”, unifying various Office of Special Enforcement registration variants for consistent categorical treatment.

3.4 Target Variable Engineering

Since no ground-truth violation label exists in the dataset, we constructed the binary target variable `is_illegal` from first-principles legal logic, operationalizing LL18 through two criteria:

- **Criterion 1 — No License, Short-term Rental:** `license` is NaN and `minimum_nights` < 30.

- **Criterion 2 — Entire Unit, Absent Host:** `room_type` is “Entire home/apt”, `minimum_nights` < 30, and the host's neighbourhood differs from the listing's neighbourhood, with our assumption that a host whose location does not match the listing is unlikely to be physically present during the guest's stay as required by LL18.

A listing is labeled `is_illegal` = 1 if either criterion is met, yielding 1,080 illegal listings (3.0%) and 35,031 surface-compliant listings (97.0%). The class imbalance was addressed during modeling through balanced class weights and, for the MLP, explicit oversampling.

It is critical to acknowledge that this label is a conservative lower bound on actual non-compliance. Many hosts in the compliant pool still likely operate illegally. Uncovering these hidden violators, namely the “Shadow Listings”, is precisely the function of the predictive model described in the following sections.

4. Exploratory Data Analysis

4.1 Class Distribution and Analytical Approach

The target variable exhibits severe class imbalance, with illegal listings comprising 3.0% of the dataset. This imbalance is realistic: many non-compliant hosts actively obscure their status by adjusting listing parameters, and the dataset's population includes a large number of genuinely compliant private room hosts whose rentals fall outside LL18's primary scope. Because standard correlation coefficients are diluted by dominant majority-class signals in imbalanced datasets, our EDA focuses on distributional comparisons between the illegal and compliant groups using proportion charts and boxplots, which provide more informative feature-level discrimination.

4.2 Categorical Feature Analysis

In highly imbalanced datasets where only 3% of listings are labeled illegal, raw count plots are dominated by the majority class and obscure meaningful patterns. Standardized stacked bar charts are used instead, showing the proportion of compliant vs. illegal listings within each category, making even small differences in violation rates visually detectable across groups.

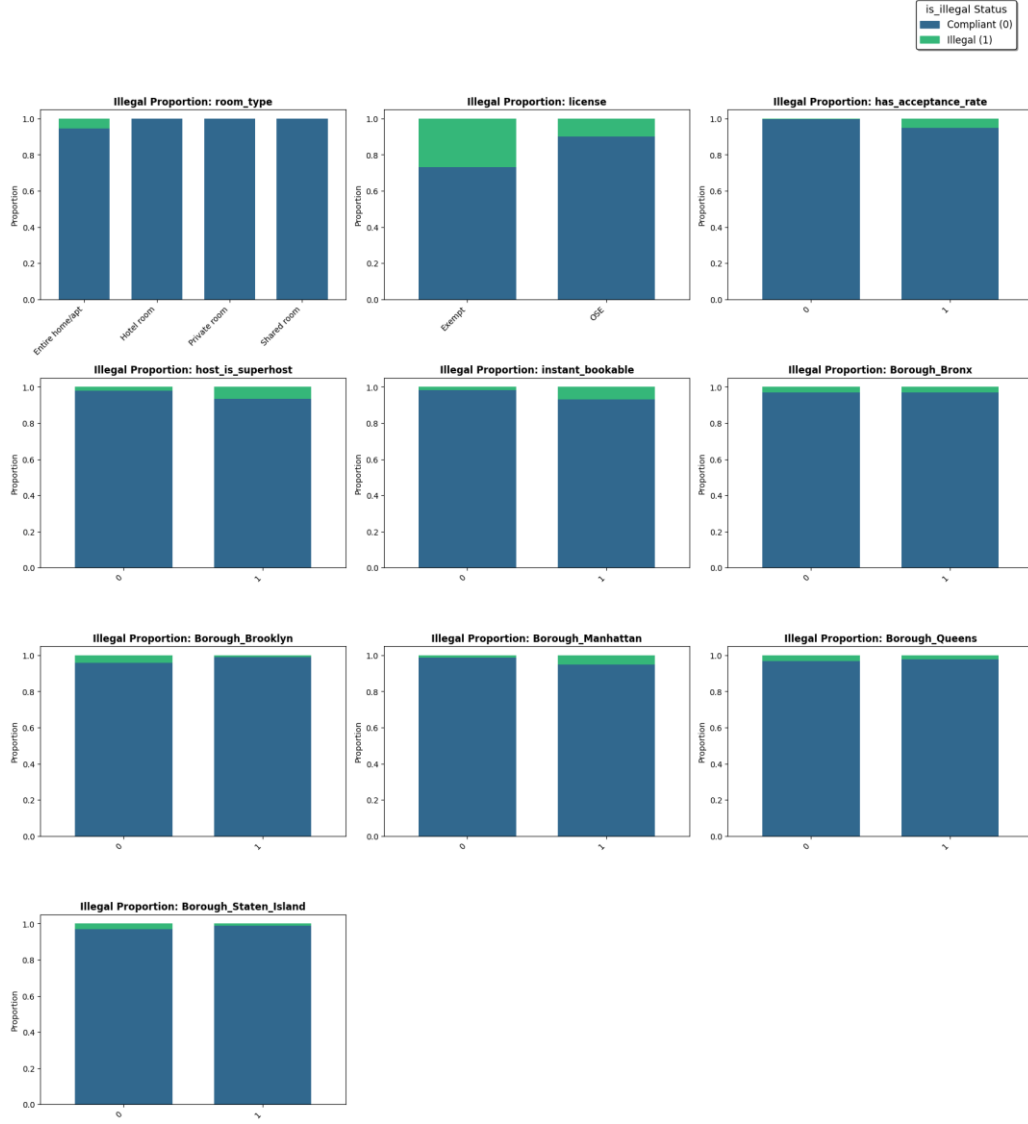


Figure 1: Categorical Feature Distributions by Compliance Status

Several features show clear separation. The `license` shows a notable pattern: Exempt listings have a substantially higher illegal proportion compared to OSE-registered listings, suggesting that the Exempt classification, which bypasses the full registration process, is associated with higher non-compliance risk. The `has_acceptance_rate` shows a relatively slight difference, with disclosing hosts carrying a marginally higher illegal proportion. Features like `host_is_superhost` and `instant_bookable` also exhibit some difference in the proportion of violators. The remaining features, namely borough dummy variables, show virtually minimal separation.

4.3 Numeric Feature Distributions

Numeric features are visualized using side-by-side boxplots, with some using log-scaled axes, as these variables exhibit heavy right skew driven by a small number of high-value listings. Log scaling compresses the extreme outliers and reveals distributional differences between the compliant and illegal groups that would otherwise be vague on a linear scale.

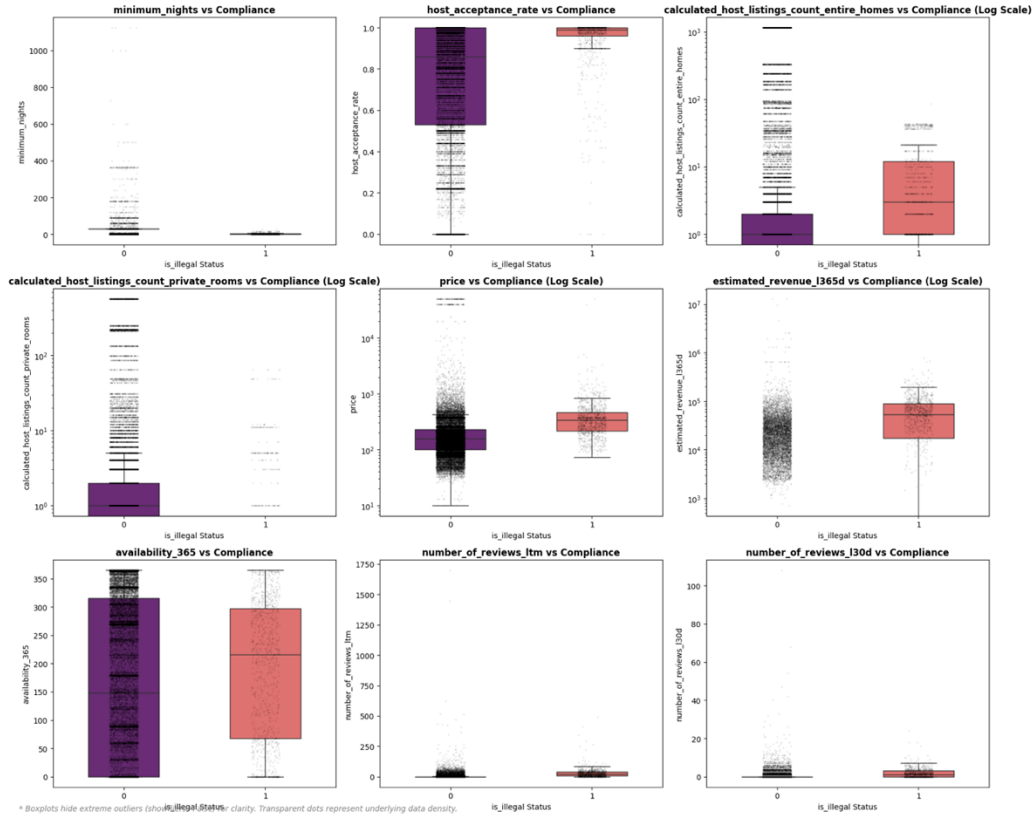


Figure 2: Numeric Feature Distributions by Compliance Status

Several features show meaningful separation between the two groups. `number_of_reviews_ltm` and `number_of_reviews_130d` are slightly higher among illegal listings, consistent with active short-term rental turnover generating frequent reviews. `estimated_revenue_1365d` and `price` are also higher in the illegal group, reflecting the premium pricing typical of non-compliance. `calculated_host_listings_count_entire_homes` is elevated among illegal listings, while `calculated_host_listings_count_private_rooms` is lower, suggesting the difference between room type. `availability_365` shows a pattern with illegal listings tending toward higher year-round availability, suggesting that illegal units tends to run as dedicated rentals rather than occasional home-sharing. `host_acceptance_rate` are also higher among illegal listings, indicating that violators are more likely to accept booking requests.

4.4 Geographic Distribution

Illegal listings are mapped using their latitude and longitude coordinates to examine the spatial distribution of non-compliance across the five boroughs.

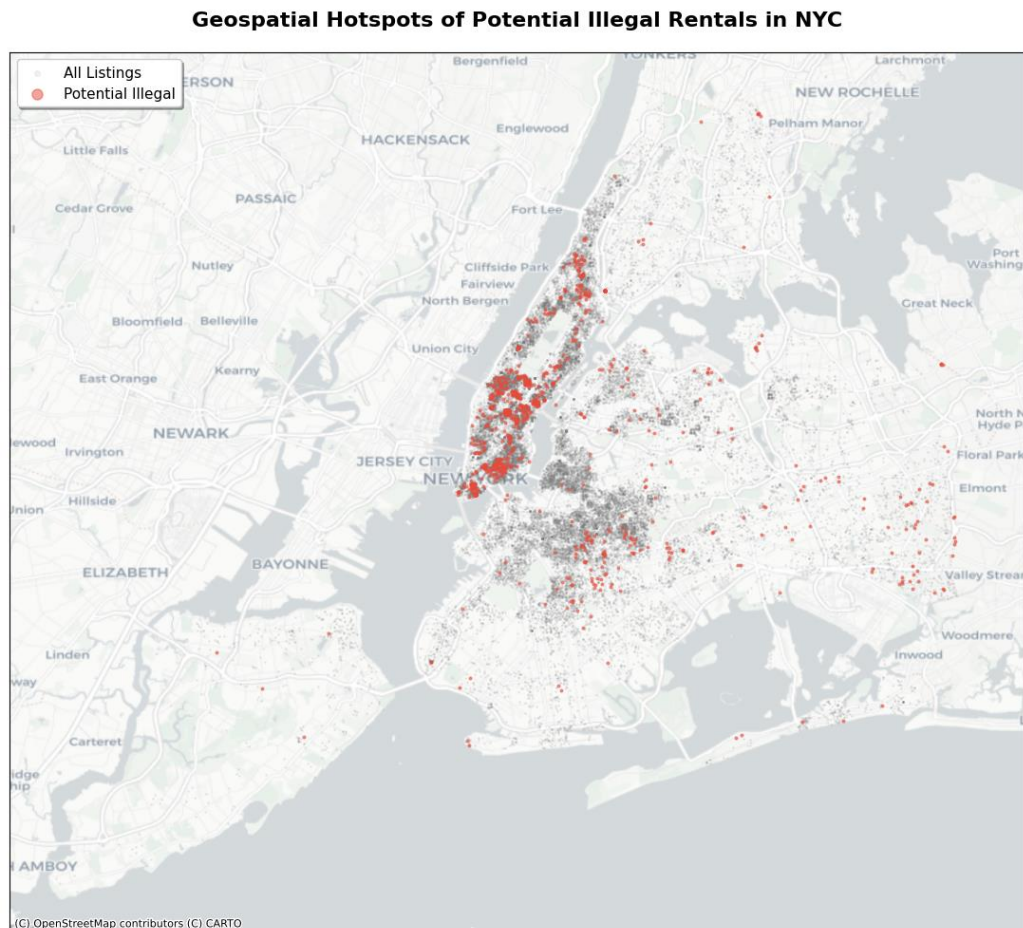


Figure 3: Geospatial Hotspots of Potential Illegal Rentals in NYC

While the borough dummy variables showed minimal separation in the categorical analysis, the geographic map reveals a clear spatial pattern. Illegal listings are heavily concentrated in Manhattan, particularly in Midtown down to Lower Manhattan, with secondary clusters in northern Brooklyn and parts of Queens. The Bronx, Staten Island, and outer Queens show comparatively sparse illegal activity. This concentration in high-demand, tourist-heavy neighborhoods is consistent with the economic incentive for illegal short-term rental operations, where premium nightly rates make non-compliance financially attractive.

4.5 Acceptance Rate v.s Superhost

This figure provides the evidence referenced in Section 3.2, justifying the differentiated imputation strategy applied to `host_acceptance_rate`.

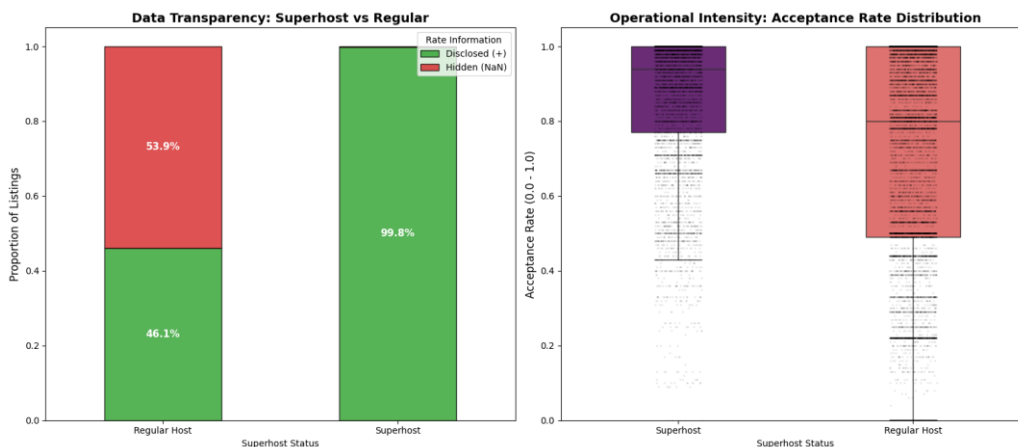


Figure 4: Acceptance Rate Disclosure Pattern and Distribution by Host Type

The left panel shows that regular hosts withhold their acceptance rate at a substantially higher rate (53.9% hidden) compared to superhosts, of whom 99.8% disclose it. The right panel confirms that the two groups differ markedly in their distributions: superhosts cluster tightly at the high end, while regular hosts show a wider spread with a lower median and more low-rate outliers. These differences justify the stratifying imputation by `host_is_superhost`.

5. Modeling

5.1 Modeling Strategy

The task is binary classification with probability outputs for each listing. Area Under the ROC Curve (AUC) on test set was adopted as the primary evaluation metric because it directly measures the model's ability to rank illegal listings above compliant ones across all possible thresholds, which is the fundamental requirement for building a useful audit priority list.

All models were evaluated on an 80/20 stratified train-test split to preserve class proportions. GridSearchCV with 5-fold cross-validation was used for hyperparameter optimization across some structured models. Class imbalance was addressed through balanced class weights (`class_weight='balanced'`) except MLP, which uses explicit upsampling with a 50/50 class ratio. The modeling pipeline proceeded in three stages: base classifiers, ensemble combinations, and supplementary text and clustering analyses.

5.2 Base Models

5.2.1 Decision Tree

Decision trees are selected for their full interpretability and robustness to outliers, as splits are based on rank order rather than absolute values, which is an advantage given the heavy right-skew and extreme values present in some features. Parameters were tuned via grid search on training data over `max_depth` (10–13) and `min_samples_leaf` (81–90) with balanced class weights, yielding `max_depth`=12 and `min_samples_leaf`=86. Test AUC: 0.9821.

5.2.2 Logistic Regression with L1 Regularization

Logistic regression with L1 penalty is included as the interpretable linear baseline. However, it is sensitive to outliers and non-linear feature relationships, which limits its performance on this dataset where several features are heavily skewed. The optimal $C=0.05$ was identified via grid search on training data, automatically dropping three features (`price`, `estimated_revenue_l365d`, `Borough_Bronx`) to zero. Test AUC: 0.9203.

5.2.3 Random Forest

Random Forest is selected for its strong predictive performance, resistance to overfitting through ensemble averaging, and robustness to outliers inherited from its tree-based structure. This makes it particularly well-suited to our dataset's skewed distributions. Parameters were tuned via grid search on training data over `n_estimators` (100, 150, 200), `max_depth` (10, 12, 14), and `min_samples_leaf` (70, 80, 90) with `balanced_subsample` class weighting, yielding `n_estimators`=200, `max_depth`=10, `min_samples_leaf`=70. Test AUC: 0.9892.

5.2.4 MLP Neural Network

The MLP is included to capture complex non-linear interactions that shallower models may miss. While neural networks can be sensitive to outliers if not properly scaled, `StandardScaler` normalization was applied to the training data to mitigate this. The model was trained on an manually upsampled balanced dataset (56,068 samples, 50/50 class ratio). Test AUC: 0.9635, with a notable gap from training AUC (0.9994) indicating moderate overfitting.

5.2.5 Support Vector Machine

SVM with RBF kernel is included as a theoretically well-suited classifier for binary classification. However, SVMs are sensitive to outliers as support vectors can be heavily

influenced by extreme values, which may explain its relatively lower performance on this dataset. Features were standardized prior to training. Test AUC: 0.9598.

5.2.6 *k*-Nearest Neighbors

kNN is included as a non-parametric benchmark requiring no distributional assumptions. However, it is sensitive to outliers as extreme feature values distort distance calculations. The optimal $k=21$ with distance weighting was identified via grid search on training data over $k \in \{3, 5, 7, 11, 15, 21\}$. Test AUC: 0.9342.

5.3 Model Comparison Summary

Model	Test AUC	Key Hyperparameters
Random Forest *	0.9892	n_estimators=200, max_depth=10, min_samples_leaf=70
Decision Tree	0.9821	max_depth=12, min_samples_leaf=86
MLP Neural Network	0.9635	Balanced upsampling, StandardScaler
SVM	0.9598	RBF kernel, StandardScaler
kNN	0.9342	k=21, distance weights
Logistic Reg. (Lasso)	0.9203	C=0.05, L1 penalty

Table 2: Base Model Performance Summary. Asterisk (*) denotes the selected deployment model.

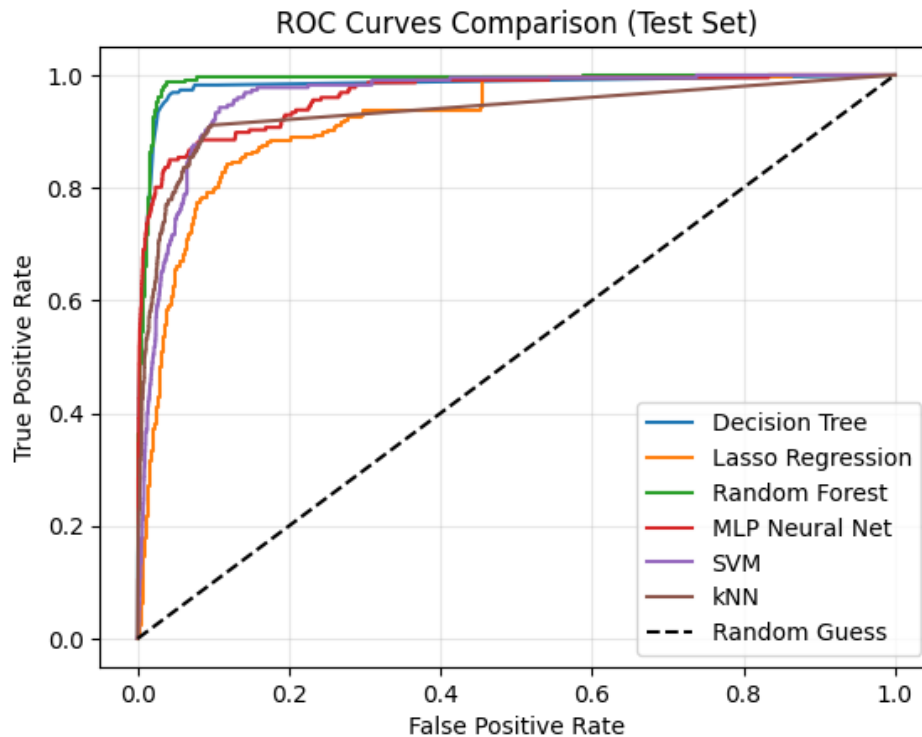


Figure 5: ROC Curves for All Six Base Models

Among all six base models, Random Forest achieves the highest test AUC of 0.9892, making it the best-performing single model.

5.4 Ensemble Methods

Following the base model evaluation, we explored ensemble combinations to determine whether aggregating predictions could yield further performance gains over the best single model.

5.4.1 Hard Voting

We first explored hard voting, where the final prediction is determined by how many models vote illegal (1), with the optimal combination and threshold identified through exhaustive search over all 63 possible model subsets and 192 situations.

Ensemble Combination	Count	Threshold	Test AUC
Random Forest	1	1	0.972785
Random Forest, MLP Neural Net	2	1	0.971925
Random Forest, SVM	2	1	0.971285
Random Forest, kNN	2	1	0.971142
Random Forest, MLP Neural Net, kNN	3	1	0.970853
...	...	...	...

Table 3: Hard Voting Ensemble Performance Summary (Top Combinations by Test AUC)

As shown in Table 3, Random Forest remains the best-performing combination under hard voting. However, hard voting produces binary class labels rather than probability scores, resulting in a notably lower AUC compared to the probability-based results reported for the base models (0.9892 vs. 0.9728). This motivated us to explore soft voting.

5.4.2 Soft Voting

Soft voting averages the predicted probabilities across model combinations, preserving more details in each model and enabling direct comparison with single-model AUC results. All 63 possible model subsets were evaluated exhaustively.

Ensemble Combination	Count	Test AUC
Decision Tree, Random Forest, MLP Neural Net	3	0.990826
Decision Tree, Random Forest, MLP Neural Net, kNN	4	0.990598
Decision Tree, Random Forest, kNN	3	0.990243

Ensemble Combination	Count	Test AUC
...	...	...
Random Forest	1	0.989189
...	...	...

Table 4: Soft Voting Ensemble Performance Summary (Top Combinations by Test AUC)

As shown in Table 4, the best-performing combination under soft voting is Decision Tree + Random Forest + MLP Neural Net with a test AUC of 0.9908, marginally outperforming Random Forest alone (0.9892). Notably, Random Forest's AUC of 0.9892 is consistent with its single-model result reported in Section 5.2, confirming that the soft voting framework produces directly comparable results with single base models.

5.5 Model Selection

Despite the soft voting ensemble achieving a marginally higher test AUC (0.9908 vs. 0.9892), **Random Forest** was selected as the deployment model for the following reasons. First, the AUC difference of 0.0016 is operationally negligible. Second, Random Forest produces a cohesive set of feature importances and is natively compatible with SHAP explainability. Third, a single model simplifies the deployment pipeline and retraining cycle.

5.6 Feature Importance and SHAP Analysis

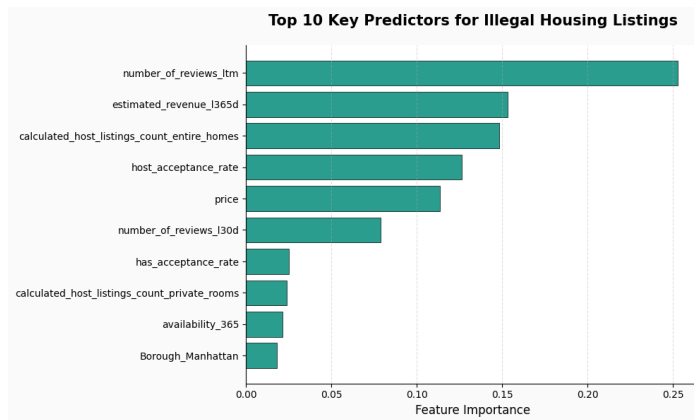


Figure 6: Random Forest Feature Importance

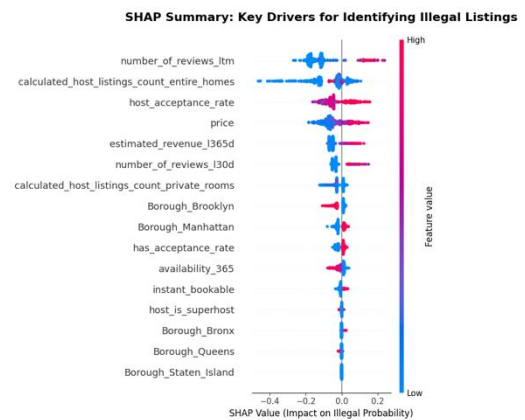


Figure 7: SHAP Summary

The feature importance chart identifies `number_of_reviews_ltm` as the dominant predictor, followed by `estimated_revenue_l365d`, `calculated_host_listings_count_entire_homes`, and `host_acceptance_rate`. This pattern is consistent with the behavioral profile of illegal commercial short-term rental operators: high guest turnover, strong revenue, management of

multiple entire-home units, and higher acceptance rate. Structural features such as borough dummies and `availability_365` contribute relatively little importance individually.

Several features show clear directional patterns in SHAP plot. High `number_of_reviews_ltm`, `host_acceptance_rate`, `price` and `estimated_revenue_1365d` consistently push predictions toward illegal, reflecting the financial motivation behind illegal commercial short-term rental operations. Some features like `calculated_host_listings_count_entire_homes` and `calculated_host_listings_count_private_rooms` don't show clear directional patterns

We notice that despite `estimated_revenue_1365d` and `price` ranking highly in feature importance (2nd and 5th respectively), Lasso regression shrinks both coefficients to exactly zero, suggesting that their relationship with `is_illegal` may not be well-captured by a linear model. To investigate this, we plot the illegal rate across binned values of both features.

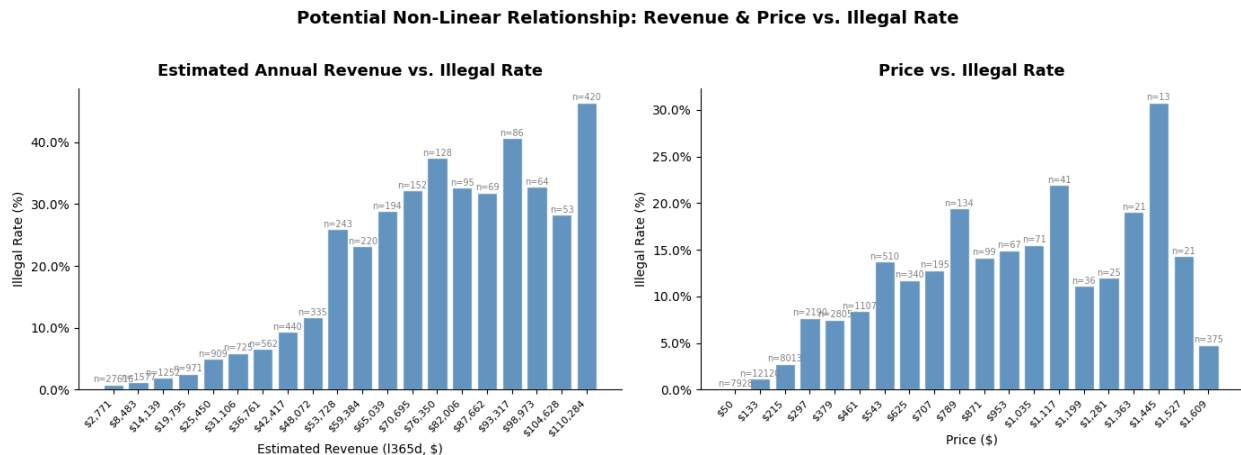


Figure 8: Potential Non-Linear Relationship: Revenue & Price vs. Illegal Rate

As shown in Figure 8, `estimated_revenue_1365d` exhibits a strongly monotonic but non-linear pattern: illegal rates remain near zero at low revenue levels and surge sharply at the high end, indicating that the signal is concentrated among a small subset of high-earning listings rather than uniformly distributed. `price` shows a more complex pattern, with illegal rates peaking at mid-range prices (~\$600–\$1,200) and declining at the extreme high end — likely because ultra-high-priced listings tend to be Class B exempt properties rather than illegal short-term rentals. Both patterns confirm that linear models cannot adequately capture these relationships, which helps explain why Random Forest substantially outperforms Lasso on this dataset.

6. Evaluation

6.1 Cost-Benefit Matrix

To translate model performance into operational ROI, we established a cost-benefit framework based on the NYC LL18 fine schedule and estimated inspection costs. Under LL18, civil penalties for hosts range from \$1,000 to \$5,000 per violation, or three times the illegally collected revenue, whichever is greater (Proper Insurance). We adopt a conservative mid-range estimate of \$2,500 per confirmed violation as fine revenue. The inspection cost of \$300 per visit is an internal estimate based on OSE inspector labor and administrative overhead, consistent with comparable municipal inspection cost benchmarks. Deducting this cost, the net benefit per True Positive is \$2,200. Each False Positive incurs only the \$300 cost with no fine revenue. False Negatives and True Negatives are assigned zero direct financial consequence, though False Negatives represent an unquantified ongoing harm to the housing market.

	Truth = Illegal	Truth = Compliant
Pred = Illegal	B(TP) = \$2,200	B(FP) = -\$300
Pred = Compliant	B(FN) = \$0	B(TN) = \$0

Table 5: Cost/Benefit Matrix (Per Audit)

Financial assumptions: fine revenue = \$2,500; inspection cost = \$300.

6.2 “True Illegal” through Monte Carlo Simulation

Since `is_illegal` is a rule-based proxy label rather than a confirmed enforcement record, and no publicly available dataset of verified illegal short-term rental outcomes exists in NYC, we cannot evaluate financial returns against true ground truth directly. To bridge this gap, we employ a Monte Carlo simulation to generate probabilistic approximations of real-world field audit outcomes.

The logic is straightforward: for each of the 35,031 surface-compliant listings, we draw a binary outcome stochastically, where the probability of being a confirmed violation equals that listing’s model-predicted `prob_illegal`. For example, a listing with `prob_illegal` = 0.8 is treated as having an 80% chance of being genuinely illegal; a listing with `prob_illegal` = 0.05 is treated as 95% likely compliant.

This approach mirrors standard practice in credit risk modeling, where Monte Carlo simulation draws binary default outcomes for each borrower using their predicted default probability,

enabling portfolio-level loss estimation in the absence of observed outcomes (Joseph; Durojaiye and Sahi).

The resulting simulated outcomes are then used to compute marginal profit per inspection and cumulative expected profit across the ranked audit list.

6.3 Probability Calibration

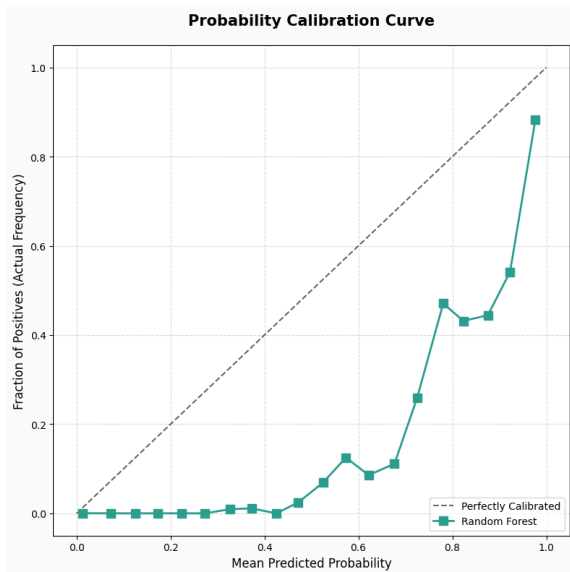


Figure 9: Probability Calibration Curve (*is_illegal*)

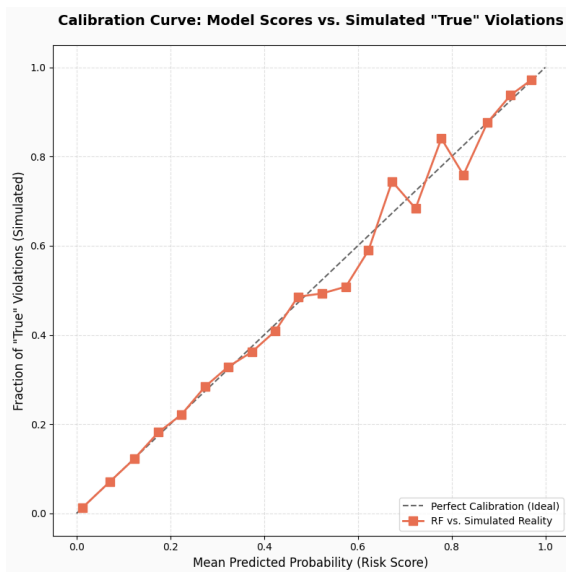


Figure 10: Probability Calibration Curve (*true_illegal*)

The first calibration curve compares the model's predicted probabilities against the observed *is_illegal* rate in the test set. The curve falls below the perfectly calibrated diagonal. In standard settings this would indicate underconfidence, but here it reflects the conservative nature of our rule-based label: many genuine violations are labeled 0 because they evade the rule-based criteria, suppressing the observed positive rate. The non-monotonic shape reflects that violations are more readily identified at the extremes of the predicted probability spectrum, while listings in the intermediate range are disproportionately under-captured.

The second calibration curve validates the Monte Carlo simulation by substituting *true_illegal* as ground truth. Its near-perfect alignment with the diagonal confirms that drawing binary outcomes proportional to *prob_illegal* produces a well-calibrated simulated dataset, lending credibility to the financial projections in Section 6.4.

6.4 Profit Curve Analysis

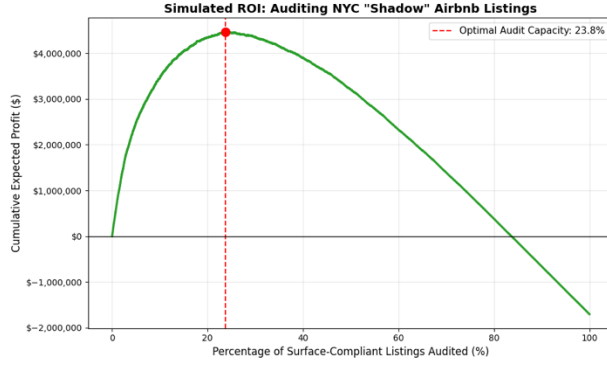


Figure 11: Cumulative Profit Curve

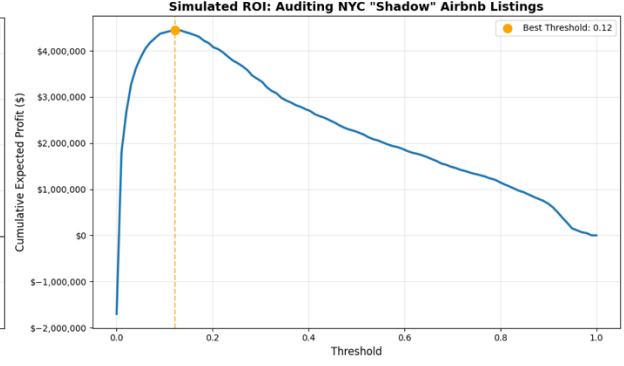


Figure 12: Net Profit by Audit Threshold

Figure 11 presents the cumulative profit curve ordered by descending `prob_illegal`, which peaks at auditing the top 23.8% of the high-risk pool (8,327 out of 35,031 listings), corresponding to a threshold of `prob_illegal` ≥ 0.1287 and an expected maximum net profit of \$4,464,400. Beyond this point, the marginal cost of inspecting lower-probability listings exceeds the expected fine revenue.

However, the cumulative profit curve alone obscures an operational nuance: the number of listings targeted does not decrease linearly as the threshold increases. Since predicted probabilities are not uniformly distributed, a small increase in threshold near the lower end can eliminate a large number of listings, while the same increment near the upper end removes very few. To make this relationship more transparent and actionable, Figure 12 plots net profit directly against the threshold value, allowing inspectors to immediately read off the financial consequence of any chosen cutoff and select the threshold that best fits their capacity constraints.

At the optimal threshold, the confusion matrix is as follows:

	Truth = Illegal	Truth = Compliant
Pred = Illegal	TP: 2,820	FP: 5,858
Pred = Compliant	FN: 703	TN: 25,650

Table 6: Confusion Matrix at Optimal Threshold

- True Positives — listings flagged as illegal and confirmed as violations by the simulation
- False Positives — listings flagged as illegal but found to be compliant
- False Negatives — genuine violations missed by the model
- True Negatives — compliant listings correctly left uninspected

The model achieves a Precision of 32.50%, a Recall of 80.05%, and an F1 Score of 46.23%. The relatively low precision reflects a deliberate trade-off: since the net benefit of catching a True Positive (\$2,200) substantially outweighs the cost of a False Positive (−\$300), the profit-optimal threshold is far below 50%.

6.5 Reality: Capacity-Constrained Scenario

In practice, OSE operates under budget and staffing constraints, making it infeasible to audit until theoretical profit peaks, namely more than 8000 inspections. We therefore simulate a capacity-constrained scenario with a fixed budget of 1,000 inspections per cycle, comparing model-driven targeting against a random auditing baseline.

Strategy	Violations Caught	Net Profit / Loss	Hit Rate
Strategy A: Random Auditing (Baseline)	91	−\$72,500	9.1%
Strategy B: Model-Driven Auditing	828	+\$1,770,000	82.8%

Table 7: ROI Comparison — 1,000-Audit Budget Scenario (Monte Carlo Simulated Outcomes).

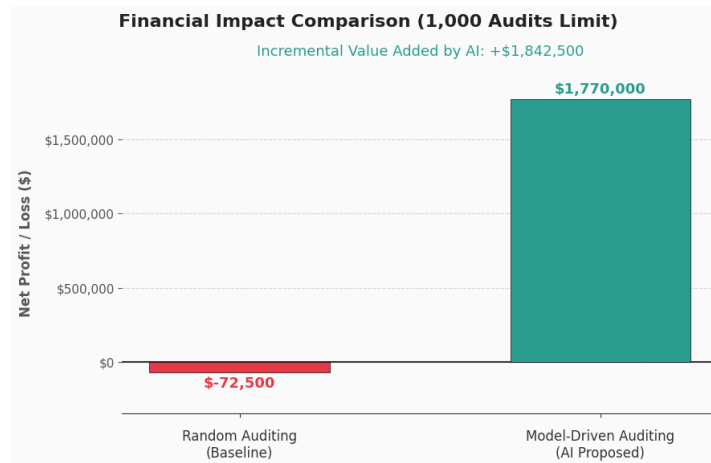


Figure 13: Financial Impact Comparison: Random vs. Model-Driven Auditing (1,000 Audits)

The model-driven strategy generates \$1,770,000 in net profit compared to a loss of \$72,500 under random auditing, representing an incremental value of \$1,842,500 attributable solely to the model’s ability to rank listings by violation probability. At a hit rate of 82.8%, the model-driven approach identifies violations at a rate more than nine times higher than random selection. This comparison provides a clear, quantified business case for adopting the risk-scoring system.

At the selected capacity, the confusion matrix is as follows:

	Truth = Illegal	Truth = Compliant
Pred = Illegal	TP: 828	FP: 172
Pred = Compliant	FN: 2695	TN: 31336

Table 8: Confusion Matrix at Selected Capacity

The model achieves a Precision of 82.80%, a Recall of 23.50%, and an F1 Score of 36.61%.

7. Supplementary Analysis

7.1 Text Analysis: Naive Bayes on Listing Descriptions

Although listing descriptions were excluded from the structured model due to their subjective and unstructured nature, we conducted a supplementary text analysis to investigate whether non-compliance could be inferred from natural language alone. Descriptions were preprocessed through lowercasing, punctuation removal, and WordNet lemmatization, then vectorized using TF-IDF with a vocabulary of 5,000 features, and trained on an 80/20 stratified split. The classifier achieved a test AUC of 0.7928 — meaningfully above chance but substantially below the structured-feature Random Forest (0.9892). The 20-point AUC gap confirms that hosts do not systematically reveal illegal intent in their descriptions, and that behavioral and structural features remain far stronger compliance signals. The top predictive words were identified using the log-probability differentials from the Multinomial Naive Bayes classifier.

Illegal Word	Illegal Diff	Legal Word	Legal Diff
jurny	5.161672	williamsburg	-2.371949
amp	4.706425	blueground	-2.233000
automated	4.677978	love	-2.086687
...	...	...	...

Table 9: Top Words Most Indicative of Illegal vs. Compliant Listings

The top words most associated with illegal listings include 'jurny' (a commercial property management platform widely used by professional operators), 'automated', and 'amp' (a rental management tool), all consistent with professionally managed, non-owner-occupied properties where the host is absent during guest stays — precisely the condition LL18 prohibits. Terms most associated with compliant listings include 'love', 'williamsburg', and 'blueground' (a platform specializing in long-term furnished rentals), suggesting more personal hosting relationships where hosts may be genuinely present and engaged with their guests.

7.2 Similarity Analysis: K-Means Clustering

To characterize the heterogeneity within the 35,031 surface-compliant listings, we applied K-Means clustering on all 16 standardized numeric features to identify distinct operational profiles within the high-risk pool.

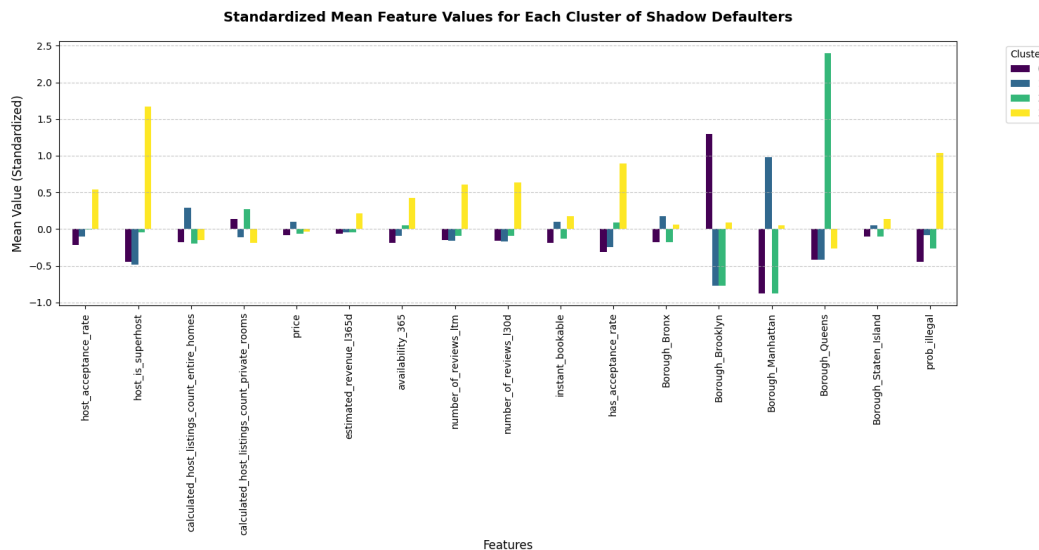


Figure 14: Standardized Mean Feature Values by Cluster ($k=4$)

Setting $k=4$ reveals a fourth distinct high-risk cluster (yellow) with an average `prob_illegal` of 0.2766, which is substantially higher than the other three clusters (0.0245, 0.0871, 0.0555), confirming that clustering captures some meaningful risk stratification. As shown in Figure 14, this cluster is characterized by high `host_acceptance_rate`, `host_is_superhost`, `availability_365`, etc. These patterns are consistent with our previous analysis.

8. Deployment and Future Work

8.1 Deployment Architecture

The model operates as a monthly scoring pipeline integrated with the Inside Airbnb data release cycle. Each month, the pipeline ingests the updated listings CSV, applies the feature engineering steps, generates risk scores via 5-fold cross-validation using our select model, and outputs a ranked list of surface-compliant listings sorted by `prob_illegal` descending. Two operational output modes are supported, namely a threshold-based list and a capacity-based list.

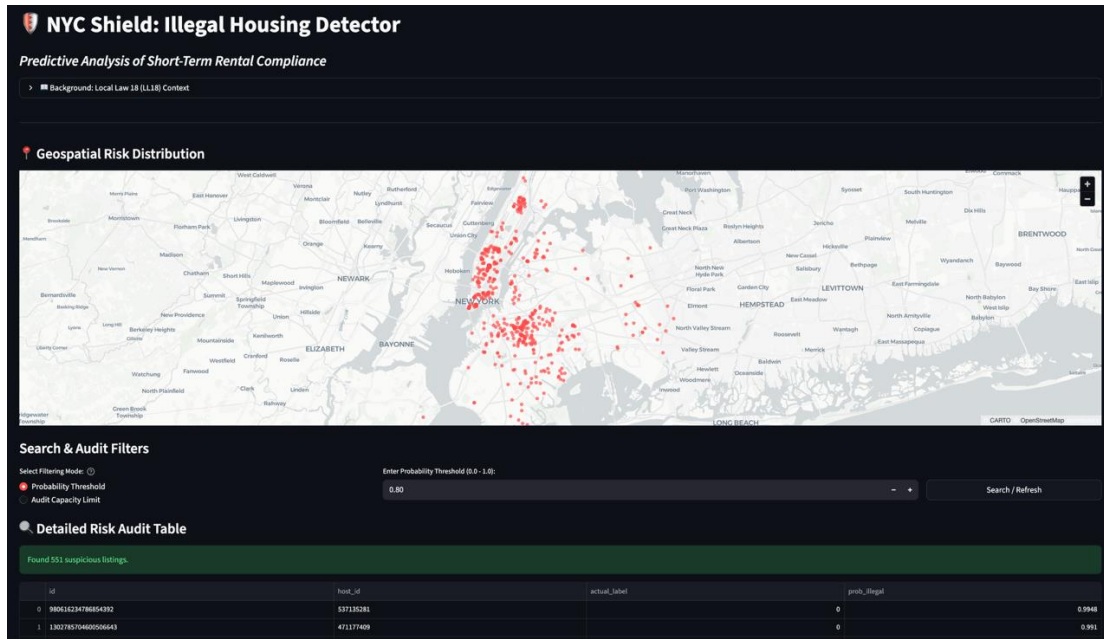


Figure 15: website prototype

To support field deployment, we developed an interactive web application called NYC Shield: Illegal Housing Detector using Streamlit, which allows OSE personnel to filter listings by threshold or audit capacity, visualize high-risk listings on an interactive map, and export a detailed risk audit table for field dispatch.

8.2 Monitoring and Model Maintenance

The model should be retrained quarterly as new Inside Airbnb snapshots are released. Two monitoring mechanisms should run in parallel: tracking the empirical hit rate of model-generated inspections against the reactive-enforcement baseline, and monitoring AUC on a rolling holdout set to detect concept drift as hosts change their listing behavior. The `is_illegal` labeling logic should also be updated whenever LL18 is amended.

8.3 Ethical Considerations

Geographic concentration of risk scores in Manhattan may create disproportionate enforcement impact in specific communities; OSE should monitor enforcement distributions by neighborhood. The proxy label introduces a risk of flagging legitimate hosts with commercial behavioral patterns, and mandatory manual review before issuing citations is essential. Finally, the scored output constitutes sensitive behavioral profiling data and should be restricted to authorized OSE personnel.

8.4 Future Work

Key extensions include: integration with NYC 311 complaint data to ground the training label in verified enforcement outcomes; linkage with Department of Finance property records to identify corporate entities managing multiple units across different host profiles; a time-series extension to detect hosts who temporarily inflate `minimum_nights` during enforcement periods and revert during peak seasons; and incorporation of the estimated social cost of missed violations using rent impact estimates from existing STR housing research, which would enable a more complete cost-benefit framework that accounts for the unquantified housing market harm of False Negatives.

9. Conclusion

This project demonstrates that machine learning can significantly improve short-term rental enforcement under NYC Local Law 18. Beginning from a dataset with no ground-truth violation labels, we constructed a binary compliance label grounded in LL18's legal requirements, engineered a 20-feature dataset, and trained six classification models as well as ensemble models. The selected Random Forest achieves a test AUC of 0.9892.

The central business finding is that model-driven auditing delivers dramatically superior ROI compared to reactive enforcement. Under a 1,000-inspection budget, the model generates \$1,770,000 in net revenue against a baseline loss of \$72,500, with an improvement of \$1,842,500. At full deployment capacity (top 23.8% of the high-risk pool), projected net revenue reaches \$4,464,400. These results demonstrate that the model's deployment is projected to generate substantial financial returns

By deploying this system as a monthly scoring pipeline, OSE can directly address the core challenge identified in Section 1 — efficiently identifying illegal operations at scale. The transition from reactive to proactive enforcement not only increases the department's inspection hit rate but also creates a meaningful deterrent effect, protecting New York City's long-term housing supply for the residents it is designed to serve.

Citation

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Appendix

A. Group Contribution

Member	Net ID	Primary Contributions
Jiaqi Zhu	jz6553	Jiaqi was responsible for Modelling and Profit Analysis, as well as compiling and organizing the whole notebook and report.
Tanzim Ali	ta2812	Tanzim took care of Feature Engineering and the Exploratory Data Analysis (EDA).
Josh Kwon	bk2351	Josh was in charge of writing the Results, Impact, and Future Work sections.
Ziyu Tian	zt2382	Ziyu was in charge of the Business Understanding and Data Understanding sections.

Table A1: Group Contribution Summary.

B. Data Source & Code Repository

Primary Dataset: Inside Airbnb — New York City (October 2025 snapshot).

Link: [Inside Airbnb](#)

Link to Python NoteBook: [Python NoteBook](#)

Link to Little Web: [NYC Shield: Illegal Housing Detector](#)